

Seasonal ARIMA (SARIMA)

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Introduction

Seasonal ARIMA (SARIMA) extends the standard ARIMA framework to series with periodic seasonality. The standard notation is $\text{ARIMA}(p, d, q)(P, D, Q)_s$ where the lower-case letters are non-seasonal orders, the upper-case are seasonal orders, and s is the seasonal period (12 for monthly with annual seasonality, 4 for quarterly, 24 for hourly with daily seasonality). SARIMA is the workhorse parametric forecasting tool for economic, financial, and many biomedical time series with stable seasonal patterns.

Prerequisites

A working understanding of ARIMA models, differencing, the backshift operator, and seasonal decomposition.

Theory

The full SARIMA equation is

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t,$$

where B is the backshift operator ($By_t = y_{t-1}$). The non-seasonal AR, differencing, and MA components compose multiplicatively with their seasonal counterparts at lag s . The model accommodates trends through differencing (d) and seasonal patterns through both seasonal differencing (D) and seasonal AR/MA terms.

The famous “airline model” $(0, 1, 1)(0, 1, 1)_{12}$ for monthly airline passengers — Box and Jenkins’s headline example — demonstrates that even highly trending and strongly seasonal series can fit into the SARIMA family with modest order.

Assumptions

The series is stationary after regular and seasonal differencing; residuals are white noise; the seasonal period s is correctly specified by the `frequency()` of the input `ts` object.

R Implementation

```
library(forecast)

x <- AirPassengers

fit <- auto.arima(log(x), seasonal = TRUE)
summary(fit)

fc <- forecast(fit, h = 24)
autoplot(fc)

checkresiduals(fit)
```

Output & Results

`auto.arima()` searches over candidate orders using AICc and returns the chosen SARIMA model with its coefficients and standard errors. `forecast()` produces point forecasts and 80 % / 95 % prediction intervals. `checkresiduals()` reports the Ljung-Box test, ACF, and time-series plot of residuals.

Interpretation

A reporting sentence: “`auto.arima` on log-transformed AirPassengers selected an ARIMA(0, 1, 1)(0, 1, 1)₁₂ structure (AICc = -483, $\sigma^2 = 0.0014$); residuals passed Ljung-Box at lag 24 ($p = 0.61$). Forecasts for the next 24 months show continued multiplicative growth and stable seasonal pattern, with 95 % prediction intervals widening from $\pm 5\%$ at 1 month to $\pm 18\%$ at 24 months.” Always check residual diagnostics before reporting forecasts.

Practical Tips

- Use `auto.arima()` for automatic order selection; double-check the result against ACF/PACF inspection of the differenced series.
- Log-transform for multiplicative seasonality (where seasonal amplitude grows with the level); SARIMA on the log scale is then additive and well-behaved.
- For very long seasonal periods ($s \geq 52$), SARIMA becomes computationally heavy; consider regression with Fourier seasonal terms (`forecast::tslm` with `fourier()`) instead.
- Always use chronological train/test splits; random CV is invalid for time series because it leaks future information into past folds.
- Box-Jenkins residual diagnostics (Ljung-Box, ACF, PACF, normality plot) are essential after fitting; `checkresiduals()` aggregates them.
- For multiple seasonalities (weekly + yearly), use TBATS (`forecast::tbats`) or `mstl()` decomposition with non-seasonal forecasting on the adjusted series.

R Packages Used

`forecast::auto.arima()` and `Arima()` for canonical SARIMA fitting and forecasting; `fable::ARIMA()` for the modern tidyverts interface; `feasts::ACF()` and `PACF()` for diagnostic visualisations; `seasonal::seas()` for X-13ARIMA-SEATS as the official-statistics alternative.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Time series basics